Lecture Nine/Ten

Creative Thinking for AI and IoT Innovations

1. Intelligent Agents → Creative Agent: Sense, Learn, Infer and Act (Consciousness) → New actions
2. Search (Intelligent and Creative Search)
3. Expanding search spaces – A machine learning perspective on creativity

Parag Kulkarni, PhD, DSc

Quick Recap

- Intelligent and creative Systems
- Creativity
- Evaluation aspects of creativity
- Thinking – cognition
- Cognitive Flexibility
- Connectionism
- Multiplying effect of connectionism
- Uncertainty
- Mind Mapping
- Agents

Creative computing

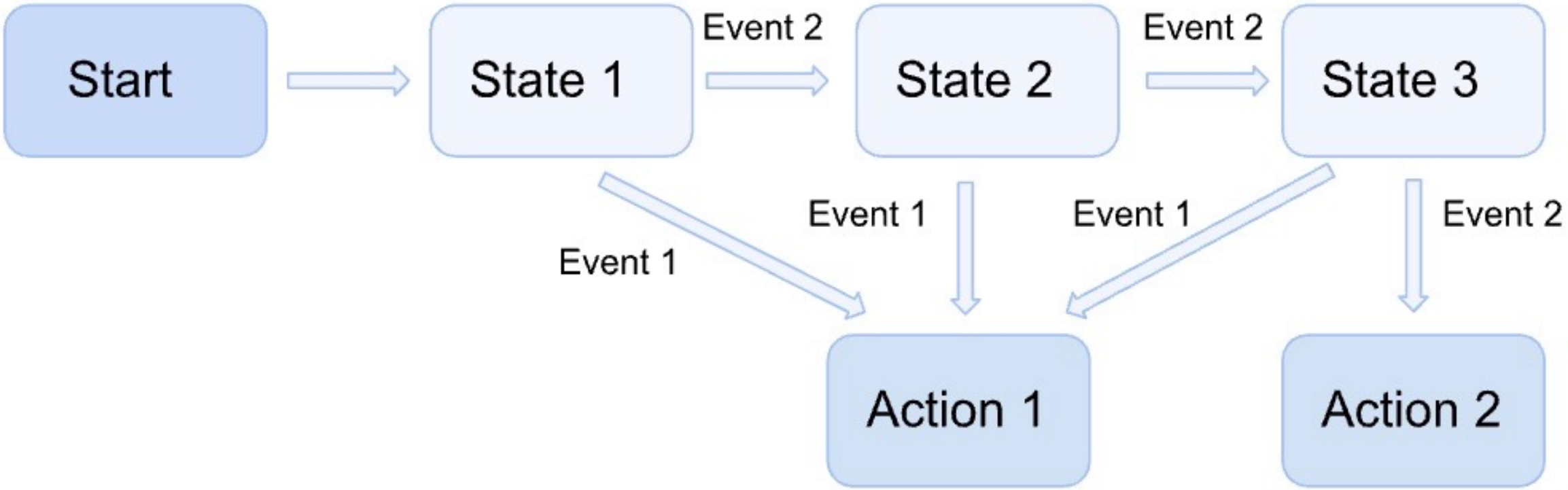
- How machines can help humans to exhibit creative behavior?
- How machines can learn from human creativity to exhibit human like creative behavior?
- How machines and humans could co-create the creative actions/products or artefacts.

- *Creativity is a process involving the generation of new ideas or concepts, or new associations between existing ideas or concepts*

  -- Wikipedia.

- Humans are creative. Ability to think out of the box.
- Goal of Computational Creativity: To model, simulate or replicate creativity using a computer.

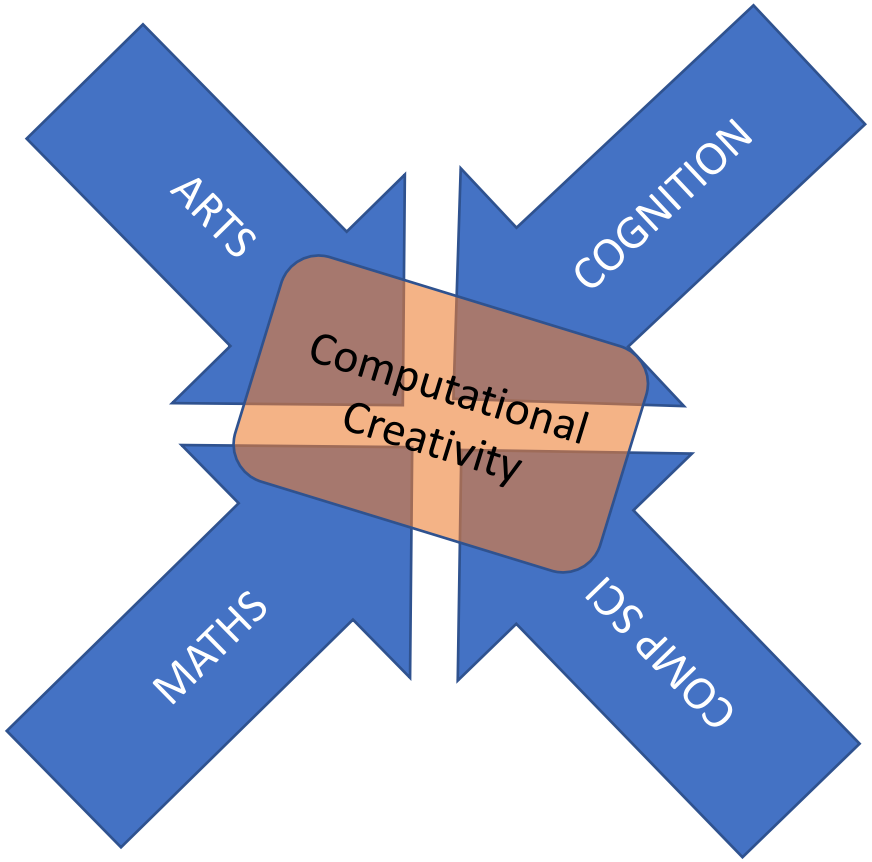
Creativity and state transition



State transition

- Introduce new action
- Introduce new state

Can you represent creativity mathematically?



CREATIVITY TEST



Machine Response



Human Responder



Creativity evaluator

Some responses may be creative!

Interrogator / investigator

Cannot detect: Who is the originator of creative response while some of them are also coming from machine

By Parag Kulkarni - Creativity and AI

Pointers with Objectives

CONCEPTS

- What is AI?
- Human Learning,
- Creativity
- Thinking
- Conceptual Spaces
- Experience and Examples
- Predictions
- Intelligence
- Learning models and Paradigms
- Building blocks of Intelligent systems
- Algorithms
- Applications
- Case-studies

PRODUCTS and SOLUTIONS

APPROACHES

- Statistical
- Associative
- Behaviorism
- Evaluation
- Evolution
- Exploration
- Genetic
- Symmetry
- Analogy
- Combination
- Time-Series Predictive
- Immune System "Group Behavior"
- Discriminative
- Generative

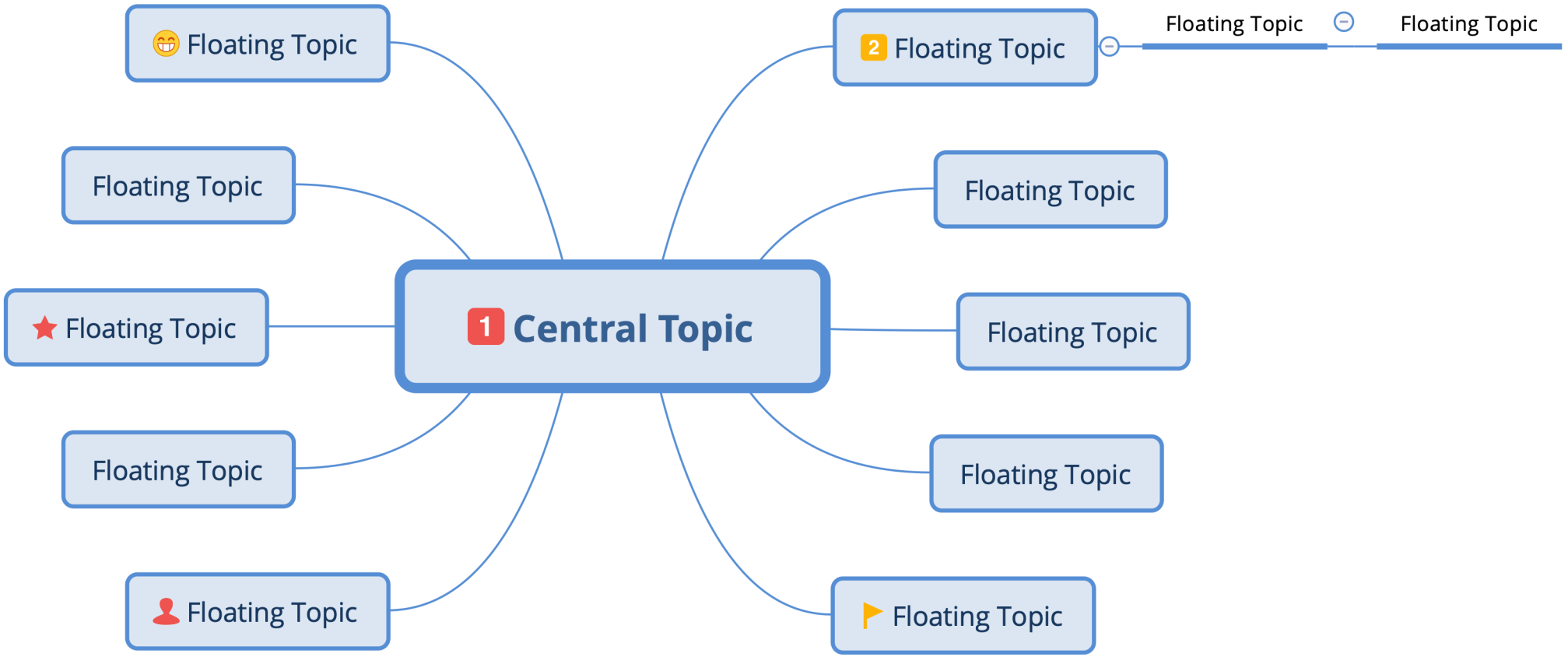
Connections and Creativity

- “Creative people tend to be open to experience, to seeing things in new ways.” - Is that about connecting?
- Facilitation to make connections
- Seeing the connection between things others could not see
- Map of connectivity and connectivity patterns
- Map of Mind: exploring connectivity to realize untapped potential
- Exploration through Possible connections and relationships
- Exploration of invisible relationships
- Creating new relationships

Mind Mapping

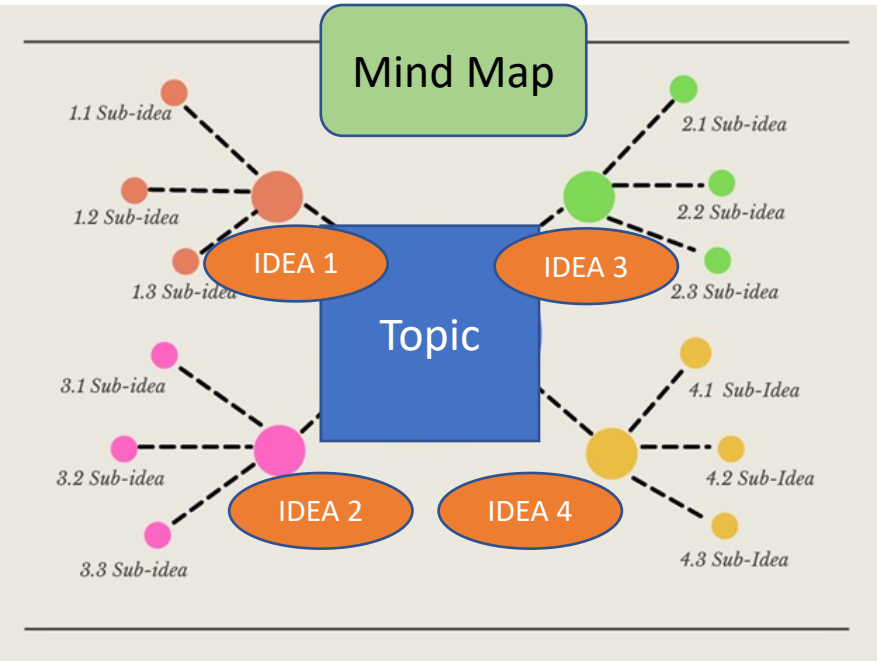
- Reinforcement through connectionism
- Beginning with random seeds and strengthening connections

X-mind and other tools

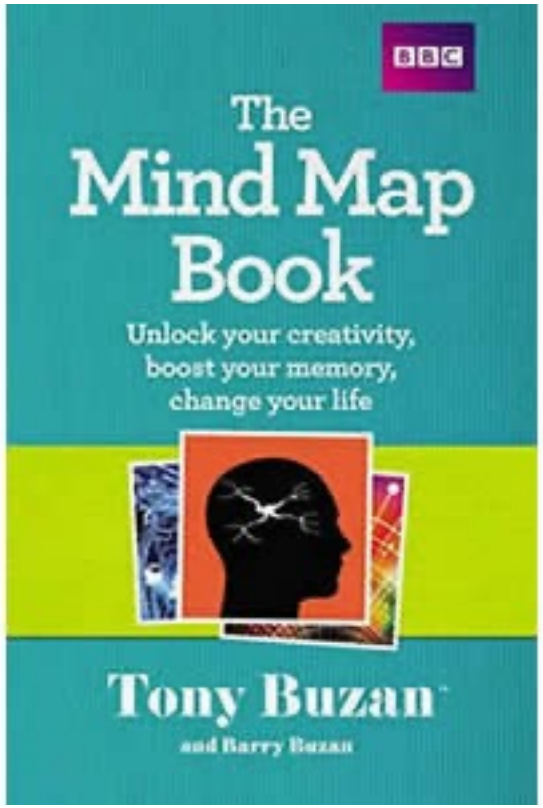

Floating Topic
Floating Topic
Floating Topic
Floating Topic
Floating Topic
Floating Topic
1 Central Topic
2 Floating Topic
Floating Topic
Floating Topic
Floating Topic
Floating Topic
Floating Topic
Floating Topic
Floating Topic

Example mind-maps

- Career
- Job selection
- Finding a route
- Product development
- Requirement gathering
- Design



More about mind mapping



Vertical Inferencing and Lateral Inferencing

- Inferencing and connections
- Imagine layers of mind maps
- Layers – association and overlap

Creative intelligence

- Opportunities to break pattern
- Strang links and patterns
- What about weak links?

Mind maps and creativity

- Application association
- Idea association
- Thinking of a big picture
- Identifying opportunities
- Product and project requirements and objectives case studies

Mind Map to creativity opportunity maps

- Unexplored relationships
- Under explored relationships
- Establishing new relationships
- Within maps and across maps

What is creative connection?

- Connection that produces association contributing to creativity
- Connections those allow to deliver useful, novel and surprising results
- Inspired from the way neurons are connected
- It can even be true when more than one agent are connected
- The basic processing system is an ensemble of such connected units.
- In a typical interconnected network, processing ends when the system has settled into a *stable state*

Reinforcement – exploration and mind maps

- Creativity driven connection reinforcement to transform mind-maps
- Developing with new starting points
- Random connection changes
- Starting with random or weakest links

Connections

- Connections as vectors
- Context based connection
- Stability not in terms of value but band
- Connection selection
- Connection based co-operation
- Evolving connections

Example

The chief enemy of creativity is good sense. —pablo picasso

- *Exploratory creativity* involves taking what is already there and exploring its outer edges, extending the limits of what is possible while remaining bound by the rules
- To understand Boden's second type, *combinational creativity*, think of an artist taking two completely different constructs and finding a way to combine them.
- *Transformational creativity* is behind those rare moments that are complete game changers.
- *Reverse hypothesis Creativity* – Beginning with the weakest hypothesis

Combinational Creativity and Creative Connections

- Combinational creativity is the process of combining old ideas to come up with something new.
- Problem-driven: Trying to bridge that gap between problem and solution by combining old ideas. an incomplete solution on their own but, once combined, prove to be a powerful solution. (Examples: ?)
- Similarity-driven: Some concepts are naturally associated inside our mind because they share similar functions, designs, and purposes. For example, a cup may make you think about a glass, a spoon about a fork, etc. (Combining two such concepts) – Example: ?
- Inspiration-driven: Connecting two or more seemingly unrelated ideas to form a new one. Seemingly random and unique, but still an example of combinational creativity. (Examples: ?)

Combination and collision can speak…

"Ideas rose in crowds; I felt them collide until pairs interlocked, so to speak, making a stable combination."

Henri Poincaré, Mathematician.

Combinational creativity and AI

- Intra/inter discipline and Cross discipline
- Idea canvas, product canvas
- Evaluation approach for combinational creativity

Creative Connection

- Can neural networks treated as a subset of creative connections
- Can it be thought of intelligent agent connection two agents/intelligent agents
- The *experiential sub-theory* connects intelligence to experience.
- Connecting agents with quite different cognitive styles.
- Connecting of disparate thoughts to form a new and relevant idea.
- Ability to see new connections and evolve them
- recording thoughts and associations through a connected nodal structure.

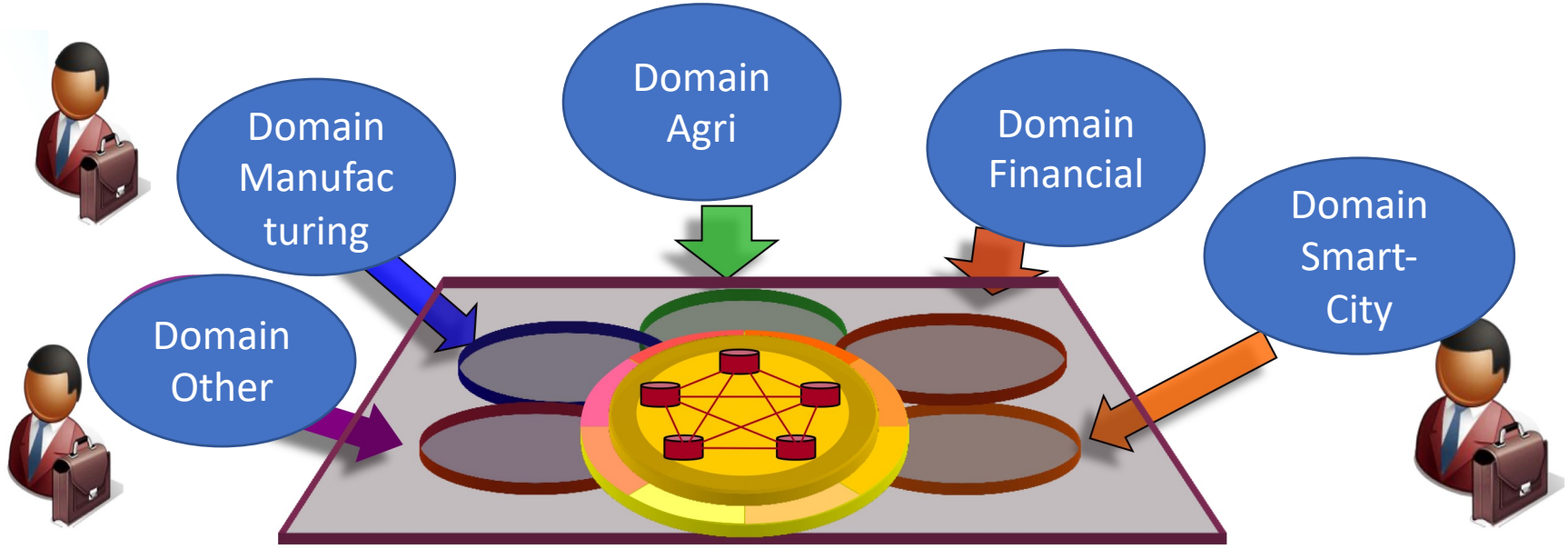
Creative Connection to build context

- the relationships among parts and between parts and the whole are viewed as interconnected, interdependent, and interactive.
- Connecting apparently unconnected concepts with creative and new connections – Here connections visualize two objects and relate them

Discovery – Creative Connections

- examine contents to discover hidden or unexpected connections or structures, either physically or, more usually in the form of recurring regularities, categories, rules, or irregularities, incongruencies, problems, and the like.
- An integrative model of affect and creativity identified the connections among affect, cognitive, and personality processes important to the creative process.
- connections may be multiple and overlapping, and involve both direct and indirect effects.

Systemic View



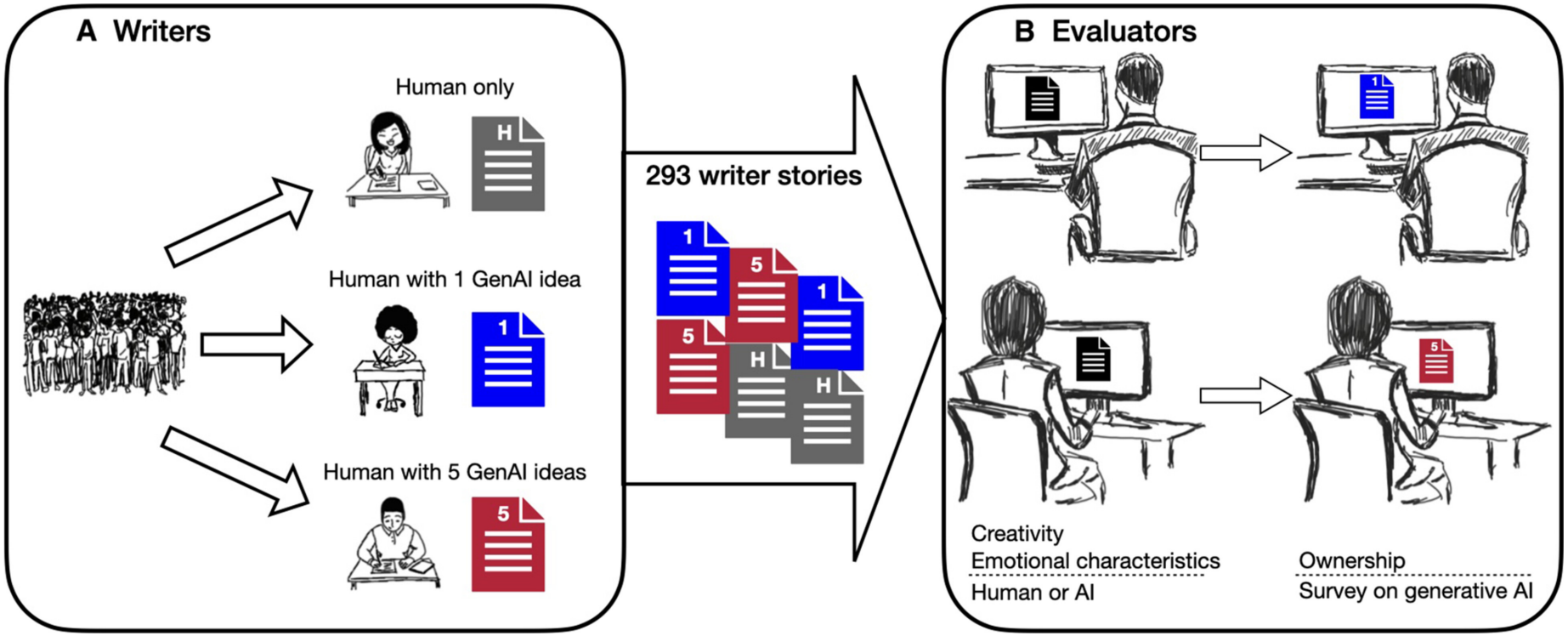
Cognitive Flexibility and creative connections

- Going beyond a single concept
- Switching importance between two or more concepts
- Relating tow or more concepts in light of execution
- Fitting different concepts in the given context of problem solving
- Adaptability to new scenario or situation

A pivot for thinking

- Let’s take a word
- What do you think about it?
- Any more word
- Word 1 + Word 2
- Add next one

Does AI enhances individual creativity but reduces the collective diversity?



Ref: https://www.science.org/doi/10.1126/sciadv.adn5290

Agents for Creativity enhancement with collective diversity inclusion

- Is decentralization in creativity is good idea
- Bad samples could be good sample
- Progress may not have right direction
- Positive + Positive < Negative + Positive
- Samples leading to questions?

Reductionism

- Reductionism sees the parts as paramount and seeks to identify the parts, understand the parts and work up from an understanding of the parts to an understanding of the whole.
- Holism considers systems to be more than the sum of their parts. It is of course interested in the parts and particularly the networks of relationships between the parts, but primarily in terms of how they give rise to and sustain in existence the new entity that is the whole whether it be a river system, an automobile, a philosophical system or a quality system. It is the whole that is seen as important and gives purpose to the study.

Reductionist Creativity, Connectionist Creativity and Inferential creativity

- Reductionist – Part at a time
- Connectionist – Find connections and association
- Inferential – Inference based on relationships

Reductionist and Connectionist Agents

Leonardo da Vinci

To develop a complete mind:

Study the science of art;

Study the art of science.

Learn how to see.

Realize that everything

connects to everything else.

Creativity and Constraints

- Process constraints and product constraints
- Expand – Reduce and Expand
- Imposing and Releasing certain constraints
- Partition based approach

Rational Agent and Creative Rational Agents

- Responsive: Respond in timely fashion to perceived environment.
- Pro-active: Should exhibit opportunistic, goal directed behavior and take the initiative wherever necessary.
- Social: Should be able to interact when they deem appropriate with other artificial agents, humans in order to compete problem solving.

Revisiting PEAS from creativity perspective

- PEAS: Performance measure, Environment, Actuators, Sensors
- Must first specify the setting for intelligent agent design
- Consider, e.g., the task of designing an automated taxi driver:
  - +Performance measure
  - +Environment
  - +Actuators
  - +Sensors

IDEA Canvas-2: Let us list different agents and see what is PEAS for them



PEAS (Crative Washing machine Vs AI Picasso)

- Must first specify the setting for intelligent agent design

- The PEAS description for it will be:
- **P**: Cleanliness of clothes, time taken, electricity consumed, water used, detergent used, noise made.
- **E**: Clothes, water tap, water, detergent
- **A**: Motor rotating the drum/ spindle, time indicator, operation completion alarm
- **S**: Water level sensor, detergent sensor, cloth weight sensor, time clock.

PEAS

- Medical diagnosis system (what are creative enhancements here)
- Auto door opening (What are creative alternatives)
- Face authentication (Any interesting addition)

Medical Diagnosis System

P - Accuracy of diagnosis, time taken, additional helpful parameters, recovery time, expenses

E - Patient, doctors, system, equipment

A - Diagnosis decision, further treatments/tests/alerts, medicines

S - BP measuring, Blood- sugar measuring, heart beat track, ECG, EEG, symptoms of patient with regard to the disease, previous reports

Auto door operating System

P - Time taken, efficiency, idle time, response time, noise made

E - Area on both sides of the door

A - Motor closing and opening the door

S - Camera on both sides, object sensing mechanism

Face Authentication System

P - Time taken, accuracy, efficiency, handling ability in case of new face

E - Face

A - Command or message indicating outcome

S – Camera/ Sensor sensing presence

Auto Car System

P - Time taken, accuracy, efficiency, safety

E - Streets, pedestrians, other vehicles, traffic signals

A - Steering wheel, accelerator, brake, horn

By Parag Kulkarni - Creativity and AI

S - Cameras, GPS signals, speedometer, engine sensors

Environments (what could be additional dimensions?)

- What are dimensions of environment?
- Fully observable (vs. partially observable): A fully observable environment is the one where the sensors of the agent can detect all aspects that are relevant. In short the agent sensors describe the environment fully or at each point the environment is entirely observable. A crossword puzzle is a fully observable environment, where as in case car system, it is partially observable.
- Deterministic (vs. stochastic): In case of deterministic environment, the next state of the environment is described completely by the current state as well as by the action of the agent. The uncertainty factor is not an issue to be looked at in case of fully observable deterministic environment.
- Discrete (vs. continuous): A discrete environment has a limited / finite number of distinct clearly defined percepts and actions. While in continuous environment those cannot be clearly defined. Chess game has a finite number of discrete states while it is not the case with football.

Environments

- Episodic (vs. sequential): In typical terms the agent's environment is divided into atomic 'episodes'. Each episode or we would call it current scene consists of agent's perception and action. So action choice in an episode wholly depends only on that episode.
- *Static (vs. dynamic):*So, we can say that environment is unchanged while an agent is deliberating. In the sense the entire scenario remains to be as it was when the agent is acting on. Vice versa is the case with dynamic environment, where the environment changes while the agent is taking action. A semi-dynamic environment is the one where the environment does not change with time slab, but the agent's performance does. In real life many examples are dynamic.

Single and multi-agent systems

- Single agent creativity
- Multi-agent creativity
- Co-operative creativity
- Ensemble creativity approach (Think of bagging and boosting)

Environment Types

	Chess with a clock	Chess without a clock	Car driving	Image analysis
Fully observable	Yes	Yes	No	**Yes**
Deterministic	Strategic	Strategic	No	**Yes**
Discrete	Yes	Yes	No	**Yes**
Episodic	No	No	No	**Yes**
Static	Semi	Yes	No	**Semi**
Single agent	**No**	**No**	**No**	**Yes**

- The environment type largely determines the agent design
- The real world is (of course) partially observable, stochastic, sequential, dynamic, continuous, multi-agent

Flexibility and Intelligent Agents

By flexible we mean that system should be:

- Responsive – respond in timely fashion to perceived environment (Tell us scenarios)
- Proactive – Should exhibit opportunistic, goal-directed behavior and take the initiative where appropriate (Temporal creativity aspects)
- Social – Able to interact, when they deem appropriate with other artificial agents, humans in order to complete problem solving (Expansion of creativity space)

Other Properties Intelligent Agent Should Have

- Mobility
- Veracity
- Benevolence – avoid conflict – do what is told
- Rationality
- learning

Structure of Agents

- Agent = Architecture + Program + (Creativity evaluator, conceptual space expander)

Sensors (current percept) → Agent Programs → Conceptual space expander → Actuators

- An agent is completely specified by the agent function mapping percept sequences to actions
- One agent function (or a small equivalence class) is rational
- Aim: find a way to implement the rational agent function concisely

Agent Environment Relationship

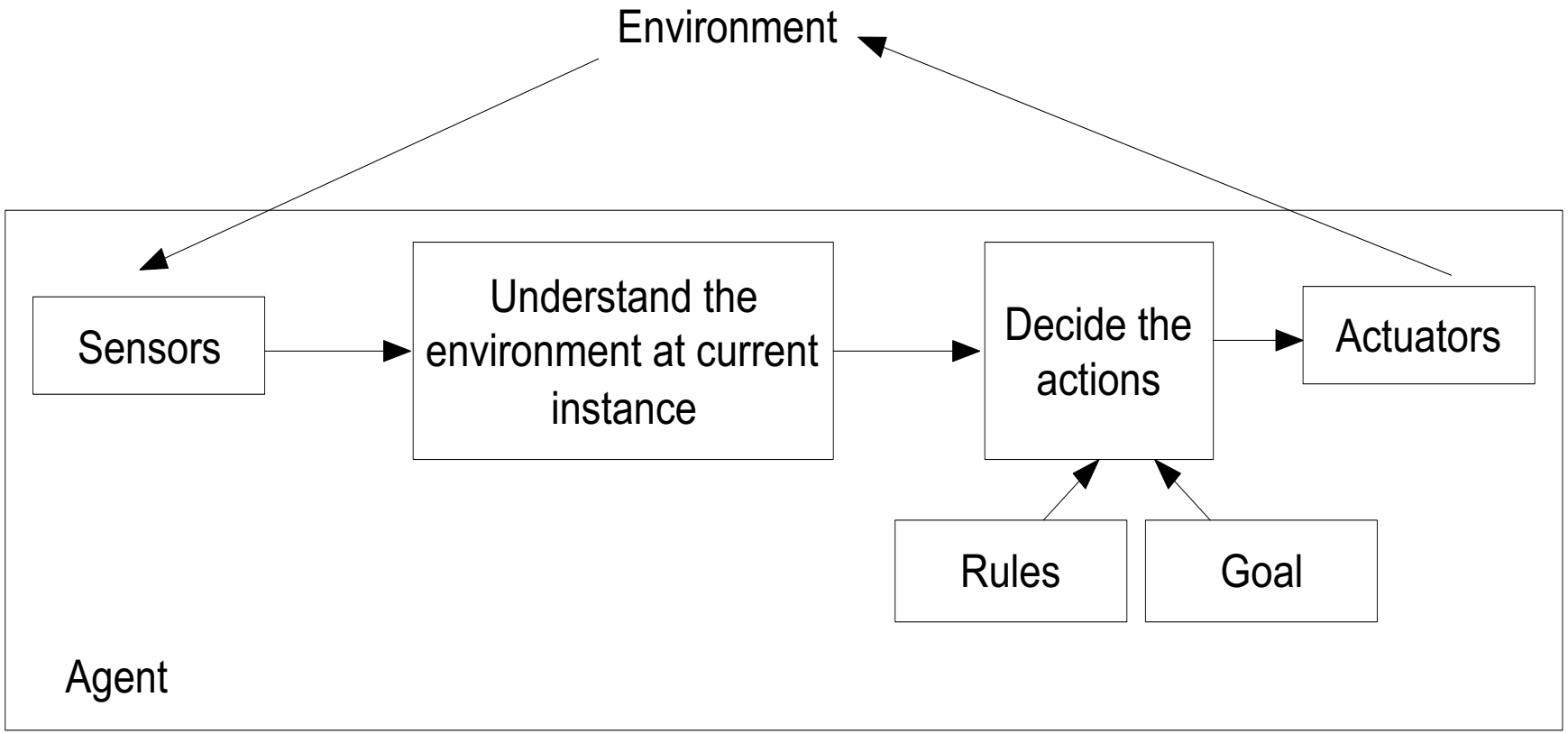


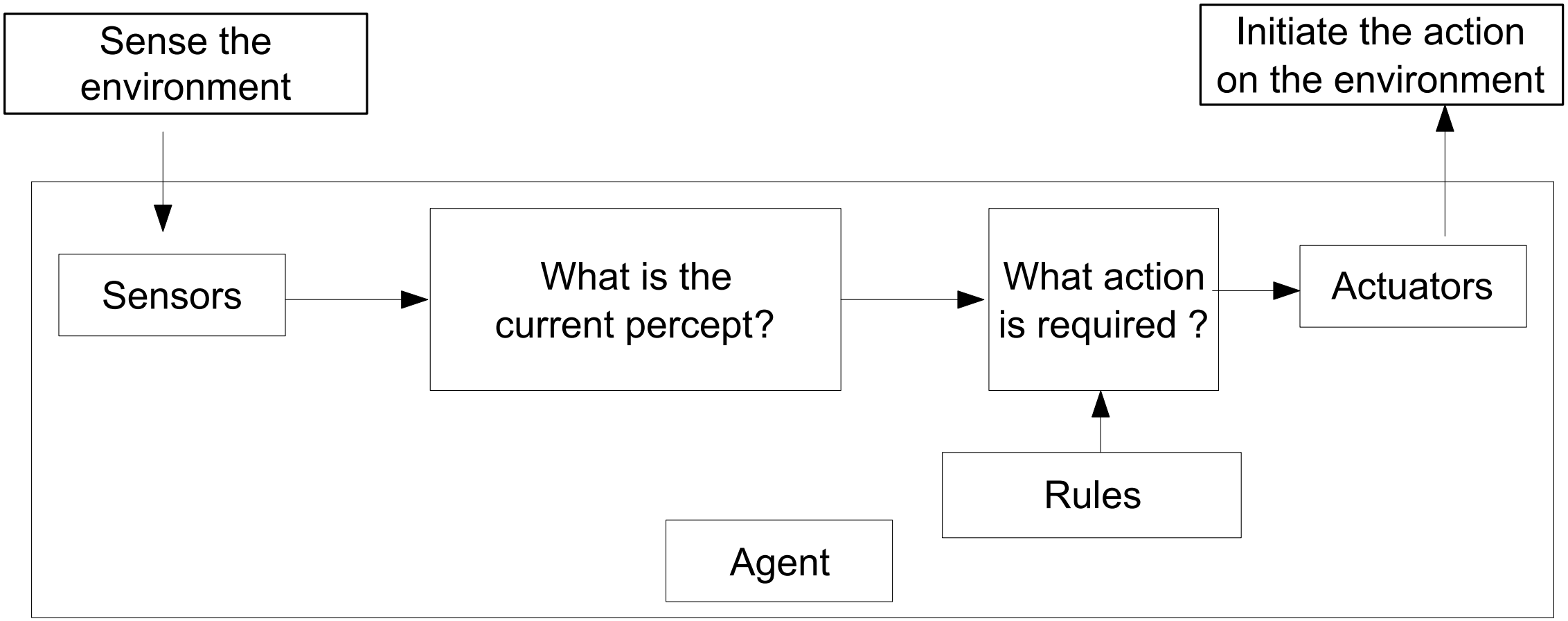
Table Driven Agents Enhancements

- A table is maintained for decision-making
- Drawbacks:
  - Huge table
  - Take a long time to build the table
  - No autonomy
  - Even with learning, need a long time to learn the table entries

Agent Types

- Simple reflex agents
- Model-based reflex agents
- Goal-based agents
- Utility-based agents

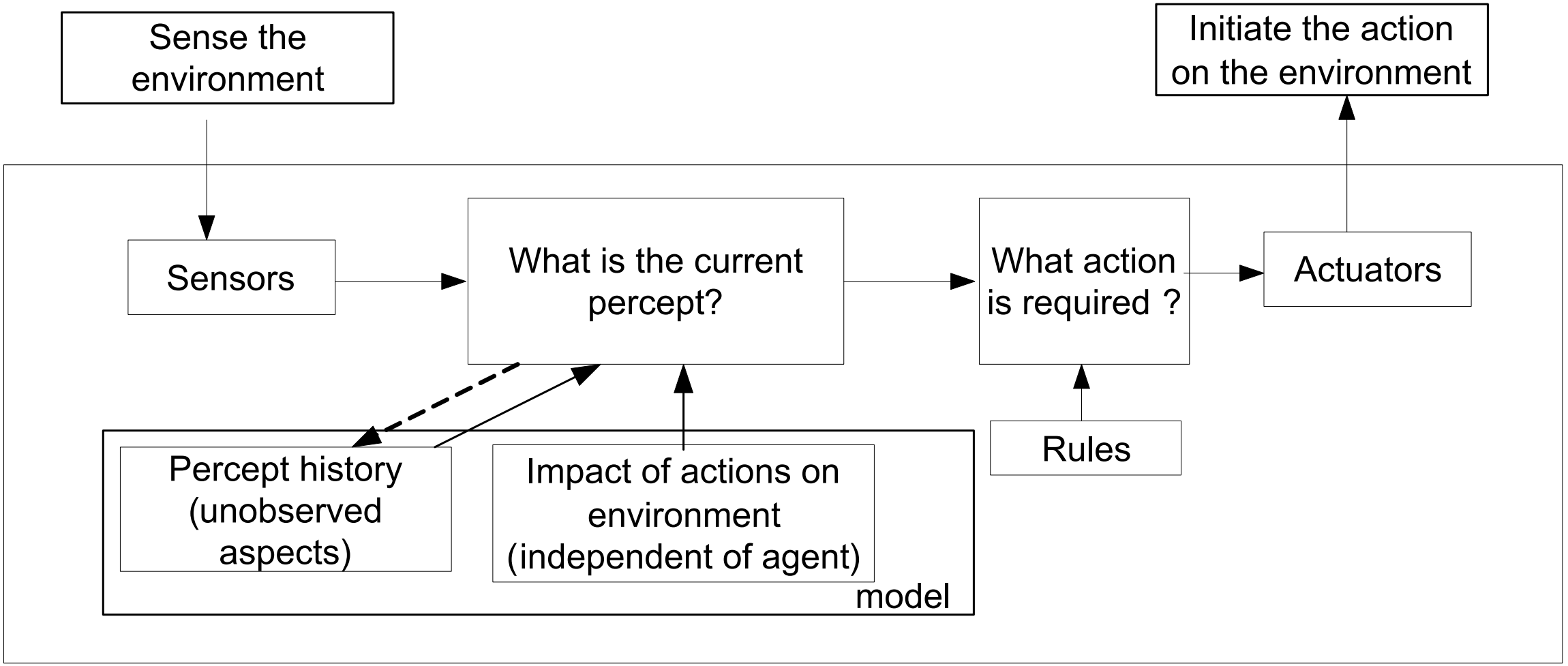
Simple Reflex Agent (Percept and creativity)



Simple Reflex Agent

- Select action on the basis of current percept (ignoring the percept history and not depend on location)
- Condition – action – rule (action driven by condition)
- Simple with limited intelligence
- Useful only in fully observable environment
- Use of randomize in case of vacuum agent

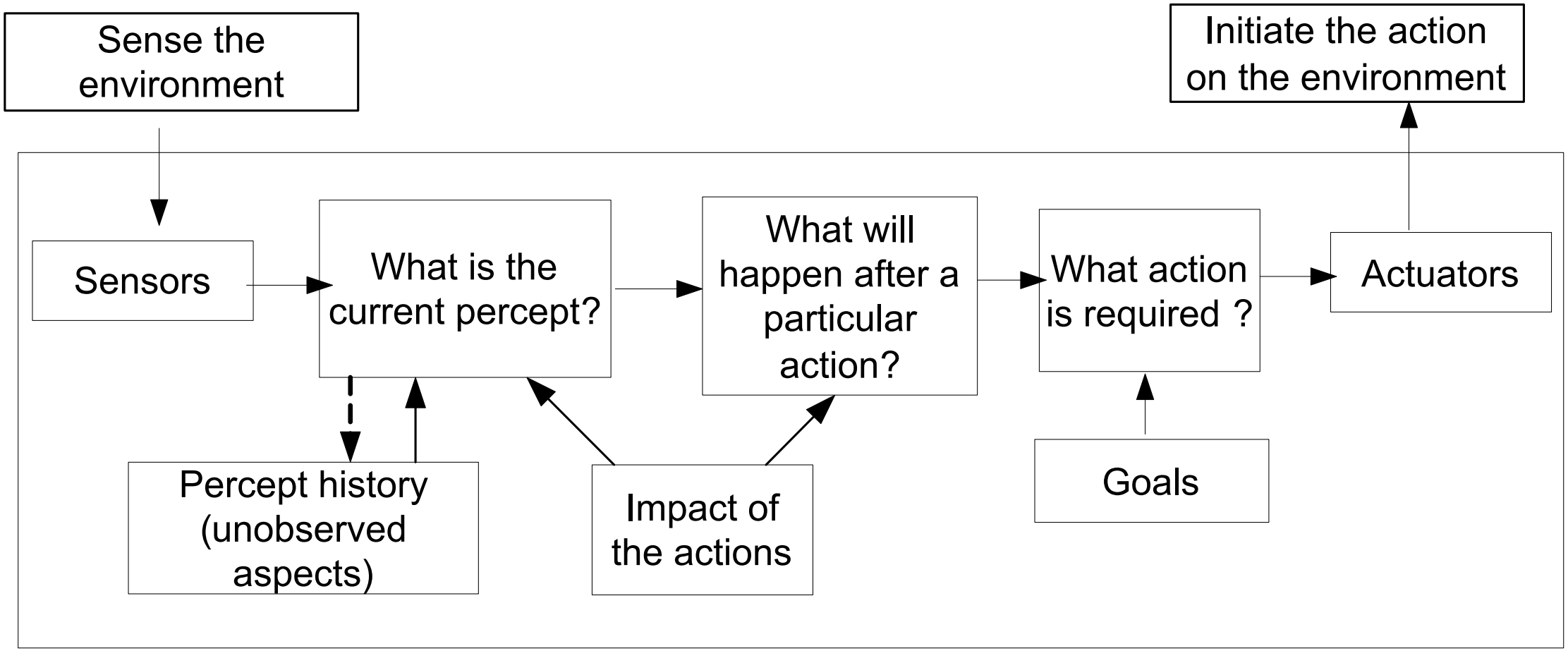
Model Based Reflex Agent (Percept sequence – Temporal Creativity)



Model Based Reflex Agent

- Keeping track of the world agent can see
- Some sort of internal state – to deal with some un observed aspects
- E.g. previous frame of camera, etc

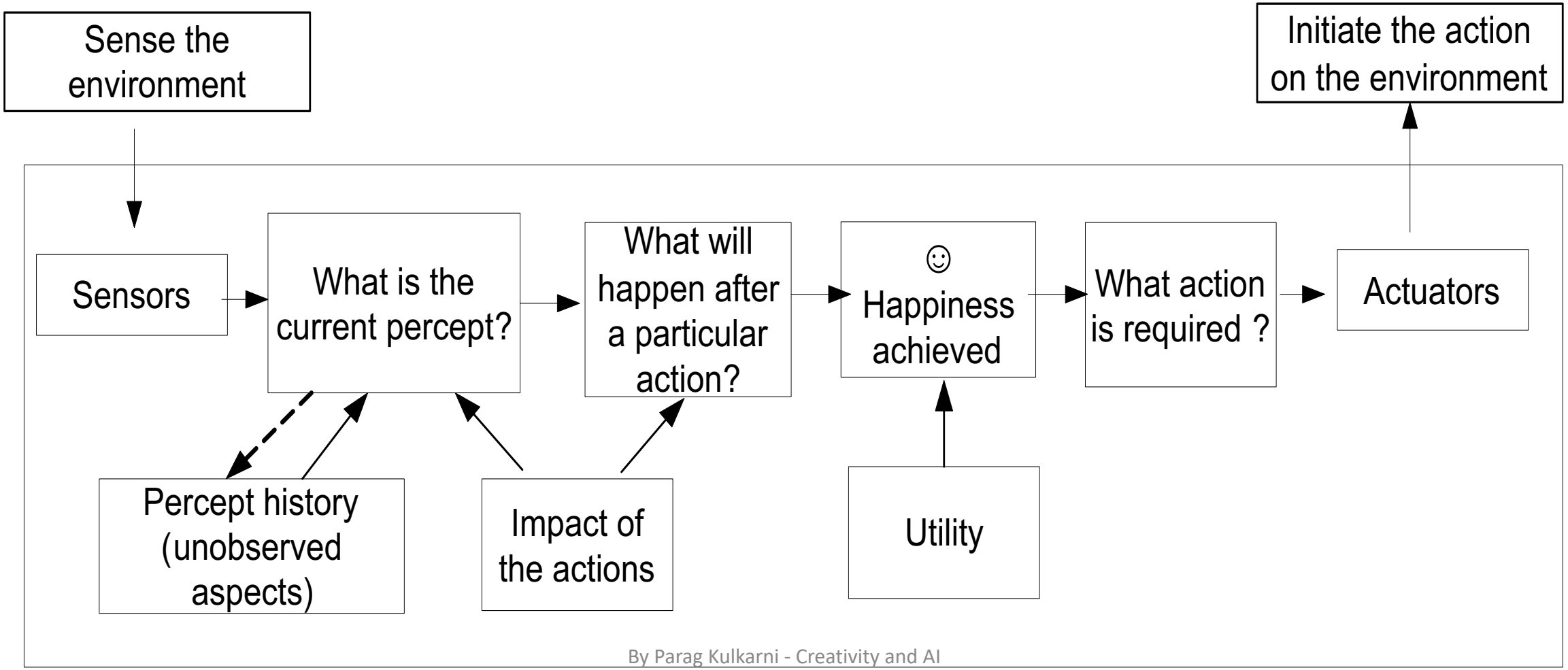
Goal-based Agents



Goal Based Agents

- Apart from the current state there is need to know the goal
- Search and planning may require for driving to goal
- More flexible as knowledge supports actions

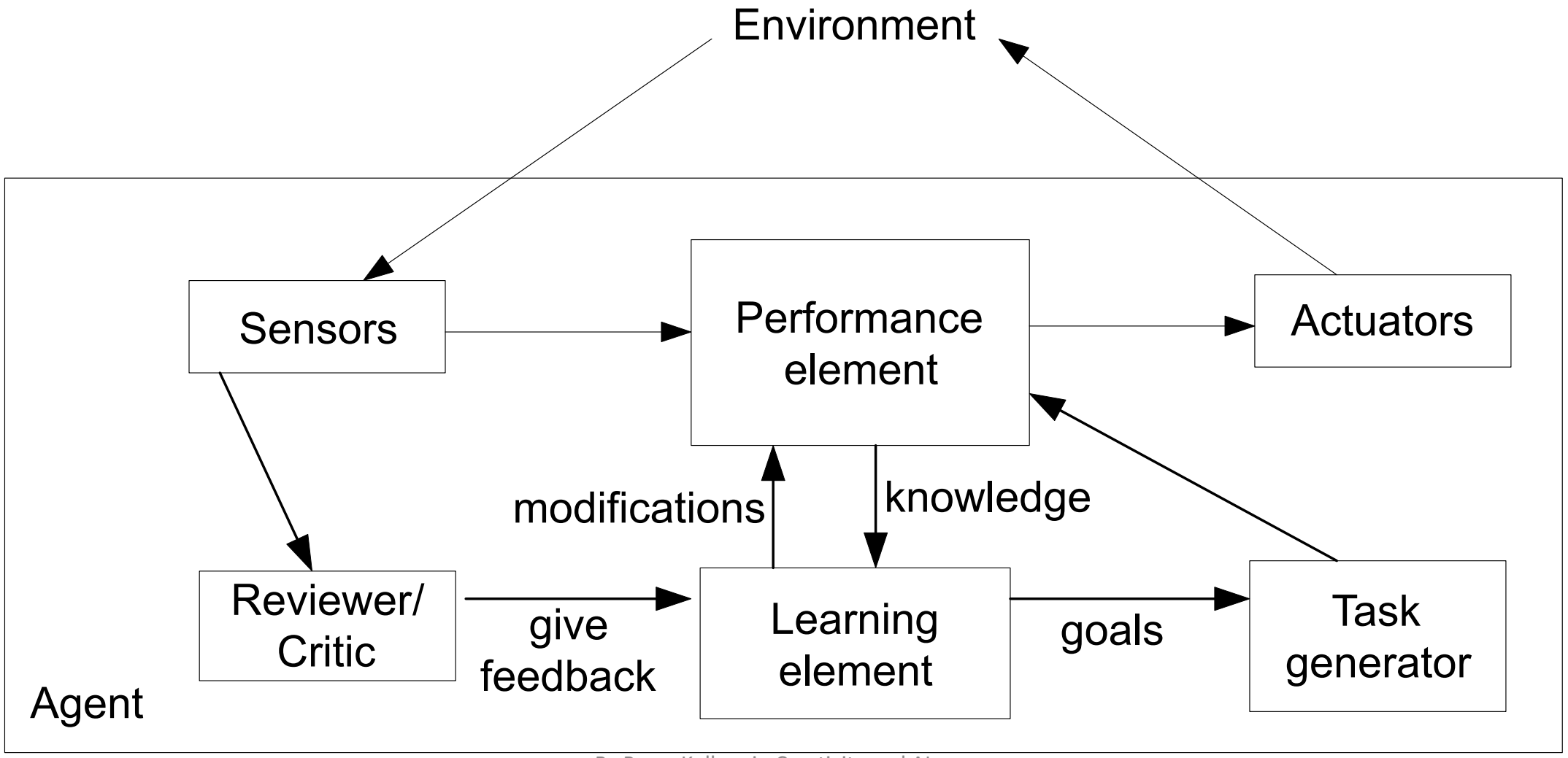
Utility Based Agents



Utility Based Agents

- Not only goal but means are important
- Some routes or series of actions are more safer, quicker, more reliable
- Happy states and unhappy states
- Utility function maps a state on to a real number which describe associated degree of happiness

Learning Agents



Learning Agents

- Ability to learn
- Learning element – responsible for making improvements
- Performance element – responsible for selecting external actions
- Learning agent uses feedback from critic
- Problem generator – responsible for suggesting actions that will lead to new and informative experiences
- Observation of behavior as a result of actions

Learning Agents

- Rewards and penalties
- Learning in an intelligent agent: process of modification of each component of agent to bring the components in to closer agreement with available feedback information – thereby improving the overall performance of the agent.

Other Systems

- Multi-agent systems
- Applications to solve problems those ate not solved so far – or solve already solved problems in better way

Creative Agent

- Function that can measure extent of creativity
- Has ability to perform obvious tasks and creative task
- Rational but non-obvious
- What could be structure of creative agent
- What type of AI problems creative agent is expected to address
- Classify tasks and outcomes on creativity scale
- Sensing previous performance – sensing competing performance

Applications

- Open systems: dynamically changing systems
- Complex systems: Abstraction
- Process control: Creative modifications
- Manufacturing: Creative designs
- Traffic control: (Ideas?)
- Information management: (Ideas?)
- E-commerce:
- Business process management
- Medical domain – monitoring, etc
- Games

Limitations Of Agent Solutions

- No overall system controllers
- No global perspective
- Creativity not possible without broader perspective

Agent Development Bottlenecks

- Requirement specifications
- System design
- System implementation
- System testing

Building Interactive Simulation Environment

- Automated pilot
- Can possible in other domains like entertainment, manufacturing and training

Think About

- Creative, dynamic and expanding Buyer agents and seller agents in E-business
- Creative Intelligent agents in network management
- Creative Intelligent agents in Business Intelligence

Searching and creativity

- Creative search strategies
- Going to state qualifying creativity criteria
- Generating new states
- Merging states
- Expand search space
- Expand action space

Talking about uninformed and informed search strategies

- Limited knowledge strategies
- Dynamic and evolving heuristic functions

Summary

- Learning Agent
- Learning Opportunities
- Adaptive learning
- Different types of learning

Lecture Ten

Creative Thinking for AI and IoT Innovations

1. Intelligent Agents → Creative Agent: Sense, Learn, Infer and Act (Consciousness) → New actions
2. Search (Intelligent and Creative Search)
3. Expanding search spaces – A machine learning perspective on creativity

Parag Kulkarni, PhD, DSc

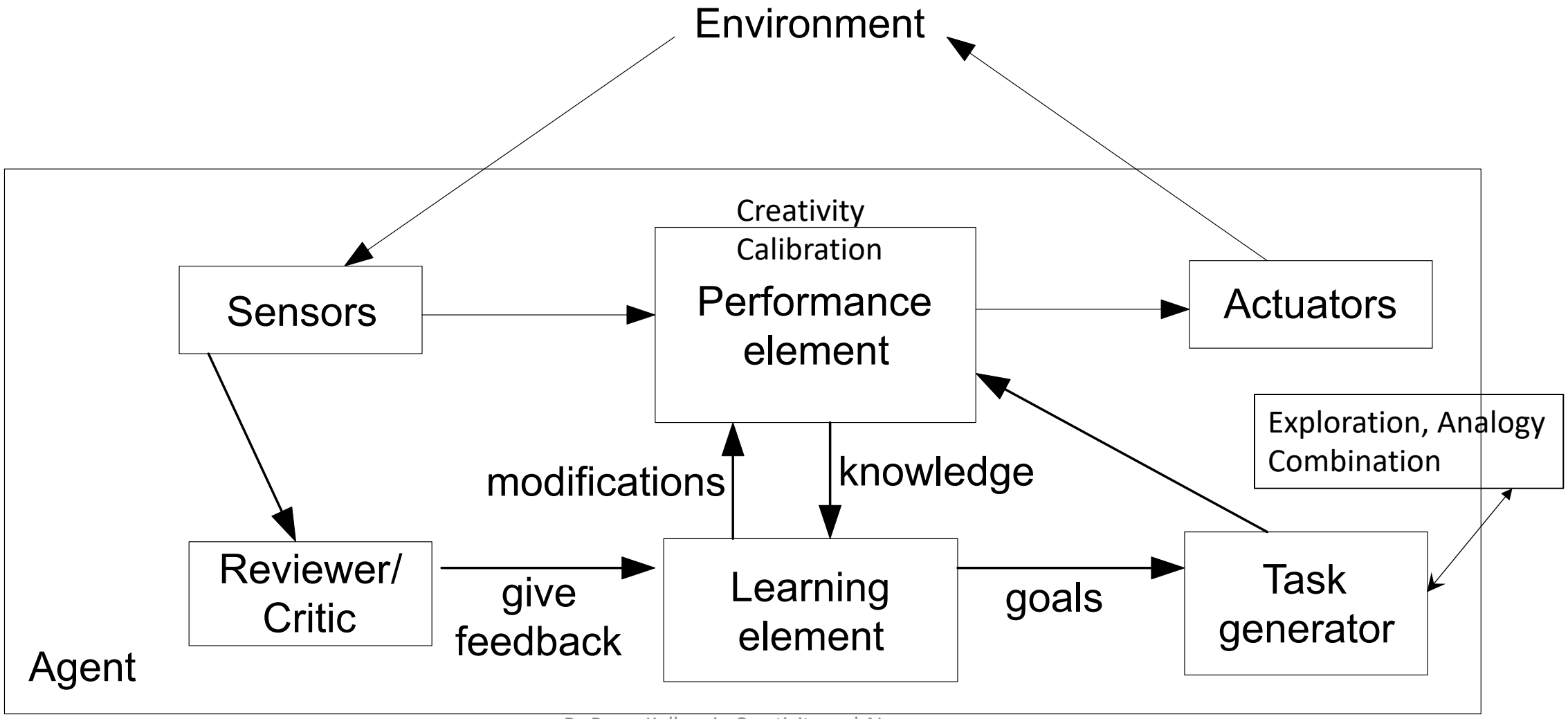
Rational Agent Should

- Responsive: Respond in timely fashion to perceived environment.
- Pro-active: Should exhibit opportunistic, goal directed behavior and take the initiative wherever necessary.
- Social: Should be able to interact when they deem appropriate with other artificial agents, humans in order to compete problem solving.

Creative Agent

- Function that can measure extent of creativity
- Has ability to perform obvious tasks and creative task
- Rational but non-obvious
- What could be structure of creative agent
- What type of AI problems creative agent is expected to address
- Classify tasks and outcomes on creativity scale
- Sensing previous performance – sensing competing performance

Creative Agents



Applications

- Open systems: dynamically changing systems
- Complex systems: Abstraction
- Process control
- Manufacturing
- Traffic control
- Information management
- E-commerce
- Business process management
- Medical domain – monitoring, etc
- Games

Limitations Of Agent Solutions

- No overall system controllers
- No global perspective

Agent Development Bottlenecks

- Requirement specifications
- System design
- System implementation
- System testing

Building Interactive Simulation Environment

- Automated pilot
- Can possible in other domains like entertainment, manufacturing and training

Talking about uninformed and informed search strategies

Searching

- Searching is a basic action
- We search problems, solutions, shops and what not
- Have you search anything recently
- What algorithm did you use
- Is it random/ad hoc/planned/organized
- What are searching applications

Searching for Solutions

- Problem can be solved by searching through the search space
- Root of search tree is a search node corresponding to initial state
- Checking whether it is a goal state
- Expanding and generating new states

Search Strategy

- The choice of which state to expand depends on search strategy
- State space and search tree
- In the theory of discrete dynamical systems, a state space is a directed graph where each possible state of a dynamical system is represented by a vertex and there is a directed edge from a to b if and only if ƒ(a) = b where the function f defines the dynamical system.

State Space

- The "state space" is the set of states of the problem we can get to by applying operators to a state of the problem to get a new state.
- The search space is the implicit tree (or graph) defined by the initial state and the operators The search tree (or graph) is the explicit tree generated during the search by the control strategy

Representations and terminologies

- State: the state in the state space to which node corresponds
- Parent node: A node in search tree that generated this node
- Action: Action that was applied to parent to generate node
- Path Cost: Path from initial state to node
- Depth: The number of steps along the path from the initial node

Fringe and leaf node

- The collection of nodes that is generated but not yet expanded is called fringe
- Each element of fringe is leaf node

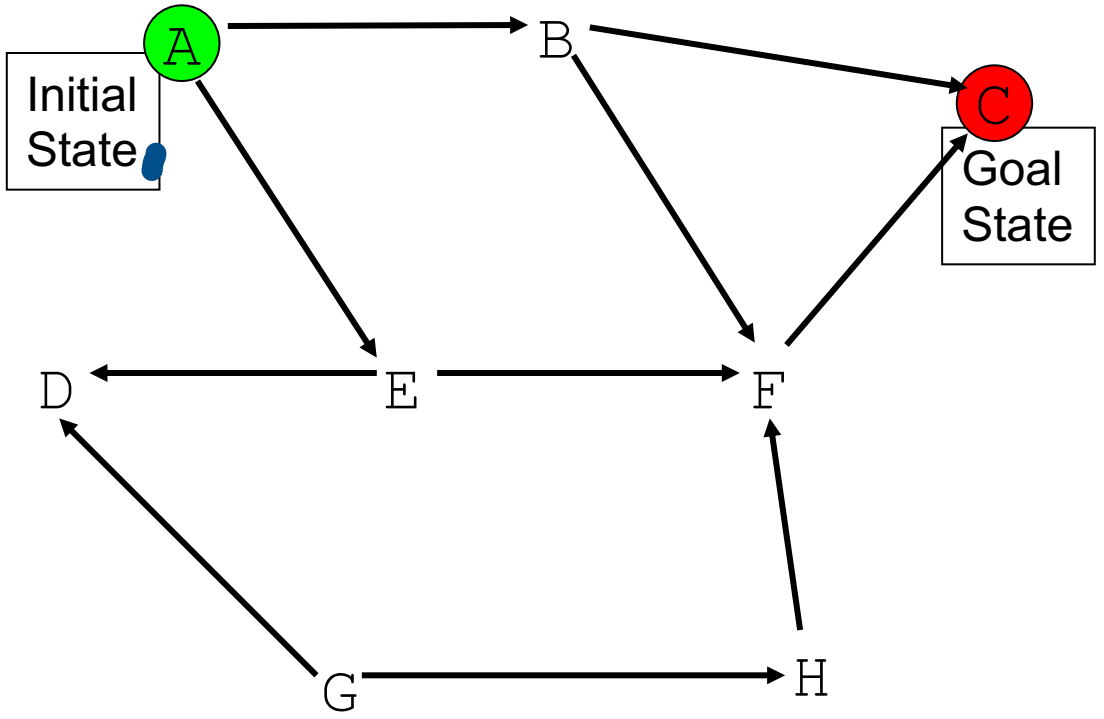
AI problems and search

- Many problems in AI take the form of state-space search.
- The states might be legal board configurations in a game, towns and cities in some sort of route map, collections of mathematical propositions, etc.
- The state-space is the configuration of the possible states and how they connect to each other e.g. the legal moves between states.
- When we don't have an algorithm which tells us definitively how to negotiate the state-space we need to search the state-space to find an optimal path from a start state to a goal state.
- We can only decide what to do (or where to go), by considering the possible moves from the current state, and trying to look ahead as far as possible. Chess, for example, is a very difficult state-space search problem.

Searching: Graph

```
link(g,h).
link(g,d).
link(e,d).
link(h,f).
link(e,f).
link(a,e).
link(a,b).
link(b,f).
link(b,c).
link(f,c).
```

State-Space




```
go(X,X,[X]).
go(X,Y,[X|T]):-
        link(X,Z),
        go(Z,Y,T).
```

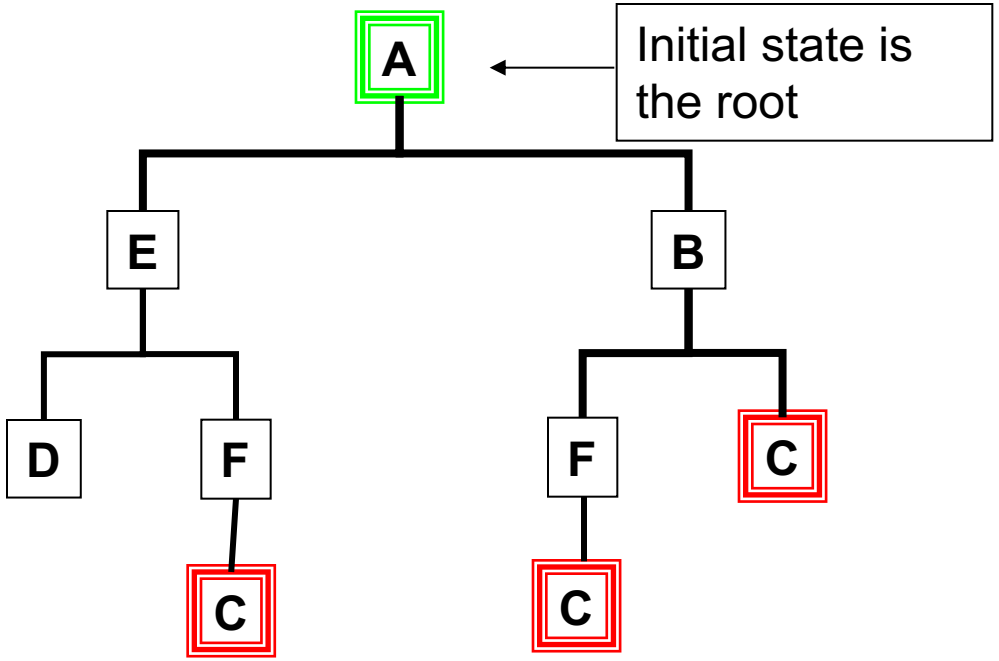
Simple search algorithm

```
| ?- go(a,c,X).
X = [a,e,f,c] ? ;
X = [a,b,f,c] ? ;
X = [a,b,c] ? ;
no
```

Consultation

STATE-SPACE REPRESENTATION

```
link(g,h).
link(g,d).
link(e,d).
link(h,f).
link(e,f).
link(a,e).
link(a,b).
link(b,f).
link(b,c).
link(f,c).
```



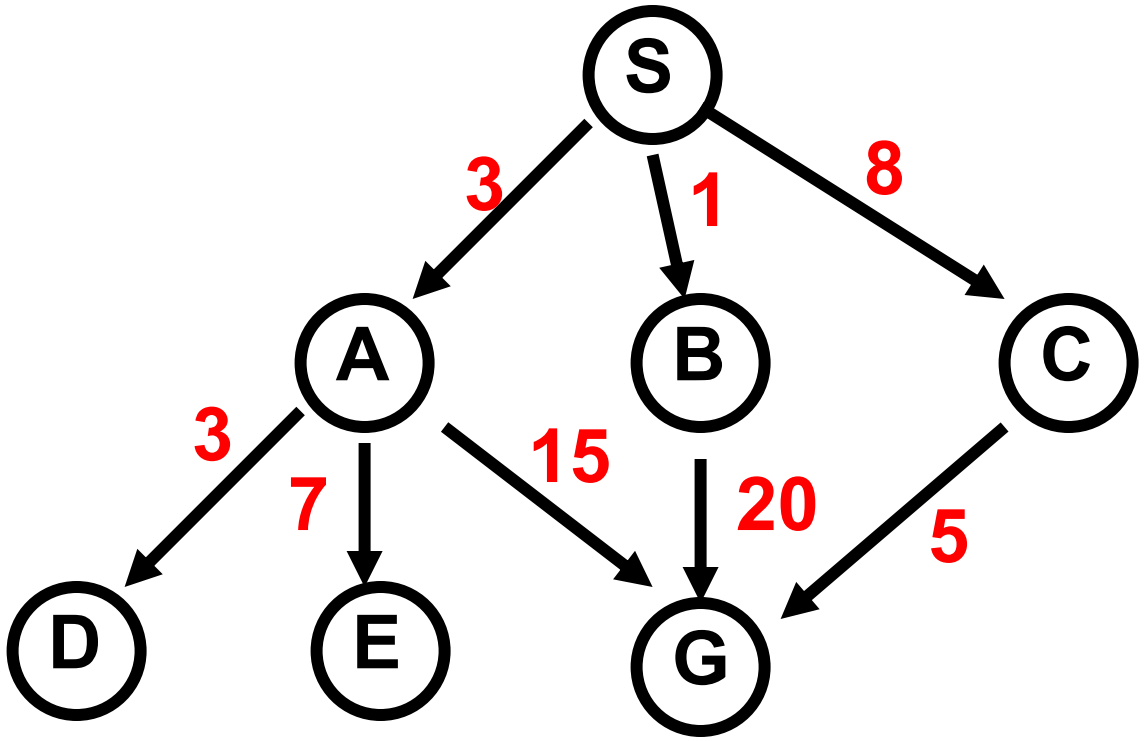
- An abstract representation of a state-space is a downwards growing tree. Connected *nodes* represent states in the domain.
- The *branching factor* denotes how many new states you can move to from any state. This problem has an average of 2.
- The *depth* of a node denotes how many moves away from the initial state it is. 'C' has two depths, 2 or 3.

By Parag Kulkarni - Creativity and AI

Uninformed Search Strategies

- **Uninformed search strategies**
  - Also known as “blind search,” uninformed search strategies use no information about the likely “direction” of the goal node(s)
  - Uninformed search methods: Breadth-first, depth-first, depth-limited, uniform-cost, depth-first iterative deepening, bidirectional
- **Informed search strategies**
  - Also known as “heuristic search,” informed search strategies use information about the domain to (try to) (usually) head in the general direction of the goal node(s)
  - Informed search methods: Hill climbing, best-first, greedy search, beam search, A, A*

Agents and environments



Evaluation of Search strategies

- **Completeness**
  - Guarantees finding a solution whenever one exists
- **Time complexity**
  - How long (worst or average case) does it take to find a solution? Usually measured in terms of the number of nodes expanded
- **Space complexity**
  - How much space is used by the algorithm? Usually measured in terms of the maximum size of the “nodes” list during the search
- **Optimality/Admissibility**
  - If a solution is found, is it guaranteed to be an optimal one? That is, is it the one with minimum cost?

Breadth First Search Algorithm

- Enqueue nodes on nodes in **FIFO** (first-in, first-out) order.
- **Complete**
- **Optimal** (i.e., admissible) if all operators have the same cost. Otherwise, not optimal but finds solution with shortest path length.
- **Exponential time and space complexity**, $O(b^d)$, where d is the depth of the solution and b is the branching factor (i.e., number of children) at each node
- Will take a **long time to find solutions** with a large number of steps because must look at all shorter length possibilities first
  - A complete search tree of depth d where each non-leaf node has b children, has a total of $1 + b + b^2 + ... + b^d = (b^{(d+1)} - 1)/(b-1)$ nodes
  - For a complete search tree of depth 12, where every node at depths 0, ..., 11 has 10 children and every node at depth 12 has 0 children, there are $1 + 10 + 100 + 1000 + ... + 10^{12} = (10^{13} - 1)/9 = O(10^{12})$ nodes in the complete search tree. If BFS expands 1000 nodes/sec and each node uses 100 bytes of storage, then BFS will take 35 years to run in the worst case, and it will use 111 terabytes of memory!

Breadth First Search

- Root node expanded first, then all successor of root node from left to right and then their successors
- Not always optimal, required more memory, may take more time

Depth First Search

- Enqueue nodes on nodes in **LIFO** (last-in, first-out) order. That is, nodes used as a stack data structure to order nodes.
- **May not terminate** without a “depth bound,” i.e., cutting off search below a fixed depth D (“depth-limited search”)
- **Not complete** (with or without cycle detection, and with or without a cutoff depth)
- **Exponential time**, $O(b^d)$, but only **linear space**, O(bd)
- Can find **long solutions quickly** if lucky (and **short solutions slowly** if unlucky!)
- When search hits a dead end, can only back up one level at a time even if the “problem” occurs because of a bad operator choice near the top of the tree. Hence, only does “chronological backtracking”

Depth First Search

- A variant is backtracking search
- Only one successor generated at a time rather than all successors
- Each partially expanded node remembers which successor to generate next
- Depth first search may make a wrong choice and may go down a very long path

Depth First VS. Breadth First

Advantages of depth-first:

- Simple to implement;
- Needs relatively small memory for storing the state-space.

Advantages of breadth-first:

- Guaranteed to find a solution (if one exists);
- Depending on the problem, can be guaranteed to find an *optimal* solution.

Disadvantages of depth-first:

- Can sometimes fail to find a solution;
- Not guaranteed to find an *optimal* solution;
- Can take a lot longer to find a solution.

Disadvantages of breadth-first:

- More complex to implement;
- Needs a lot of memory for storing the state space if the search space has a high branching factor.

Uniform Cost (UCS)

- Enqueue nodes by **path cost**. That is, let g(n) = cost of the path from the start node to the current node n. Sort nodes by increasing value of g.
- Called *"Dijkstra's Algorithm"* in the algorithms literature and similar to *"Branch and Bound Algorithm"* in operations research literature
- **Complete (*)**
- **Optimal/Admissible** (*)
- Admissibility depends on the goal test being applied *when a node is removed from the nodes list*, not when its parent node is expanded and the node is first generated
- **Exponential time and space complexity**, $O(b^d)$

Uniform Cost (UCS)

- BFS Always expand shallowest unexpanded node
- While UCS expands the node n with lowest path cost
- The completeness is guaranteed if the cost of every step is greater than zero- otherwise it may get stuck in infinite loop if it expands the node that has zero-cost action leading back to same state.

Depth Limited Search

- The typical problems encountered in depth first search due to unbounded trees can be alleviated by supplying depth first search with a predetermined depth limit.
- This solves infinite path problem
- In this case shallowest goal beyond the depth limit is missed.
- Depth first search can be viewed as a special case of depth limited search with limit infinity

Iterative deepening depth first search

- If the optimal solution is the shortest path from the initial state to the goal state depth-first search will usually not find this.
- We need to vary the depth at which we look for a solution; increasing the depth every time we have exhausted all nodes at a particular depth.
- This finds the best depth limit
- It does that by gradually increasing the limit until the goal is found
- This happens when the depth reaches d – i.e. the depth of the shallowest goal
- This technique combines advantages of breadth first search and depth first search

Iterative deepening depth first search

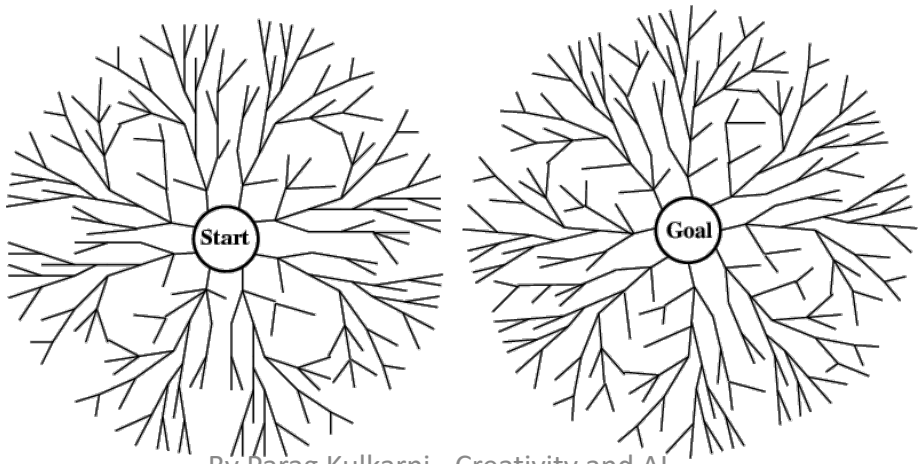
- First do DFS to depth 0 (i.e., treat start node as having no successors), then, if no solution found, do DFS to depth 1, etc.

    until solution found do
        DFS with depth cutoff c
        c = c+1

- **Complete**
- **Optimal/Admissible** if all operators have the same cost. Otherwise, not optimal but guarantees finding solution of shortest length (like BFS).
- Time complexity is a little worse than BFS or DFS because nodes near the top of the search tree are generated multiple times, but because almost all of the nodes are near the bottom of a tree, the worst case time complexity is still exponential, O(bd)

Bidirectional Search

- Alternate searching from the start state toward the goal and from the goal state toward the start.
- Stop when the frontiers intersect.
- Works well only when there are unique start and goal states.
- Requires the ability to generate “predecessor” states.
- Can (sometimes) lead to finding a solution more quickly.



Bidirectional Search

- Bidirectional search is implemented by having one or both of the searches check each node before it is expanded to see if it is in the fringe of the other search tree – if it is so then solution is found
- In this case it is required that at least one of the search tree has to be kept in the memory
- This space requirement is major weakness of bidirectional search
- The time complexity makes it attractive but faces issues in backward searching

Avoiding Repeated States

- Repeated states can make solvable problem unsolvable if algorithm fails to detect them
- Detection usually means comparing the node about to be expanded to those that have already been expanded – if it match then algorithm has discovered two paths to same state and can discard one of them
- Can help in detecting looping paths and discarding them
- It does not avoid exponential proliferation of non looping paths

Avoiding repeated states

- Tree search algorithm can be modified to include data structure called closed list - which stores every expanded node
- If the current node matches the node on the closed list – it is discarded instead being of expanded
- It can result in some disadvantages with reference to detecting optimal path in case of graph

Searching with Partial Information

- The incomplete knowledge of states and actions can lead to three distinct problem types:
  - Sensorless problems: – No sensor – Then it could be in any of the several possible initial states – each action can lead to one of the several successors
  - Contingency problems: - Percept provides new information after each action. Partially observable environment – each possible percept define a contingency that must be planned for -
  - Exploration problems: When the states and actions of the environment are unknown – agent must act to discover them

Informed Search

- Uninformed searches are about systematically generating states and testing them against goal
- Issues???
- Use of problem specific knowledge while searching

Informed (Heuristic) search strategies

- Best First Search approach
- Evaluation function
- Traditionally the node with the lowest evaluation is selected for expansion – Here evaluation function measures the distance from the goal
- All here is selection of the node that appears to be the best node
- A key component of best first search algorithms is the heuristic function h(n)

Think about applications

- Open systems: dynamically changing systems
- Complex systems: Abstraction
- Process control
- Manufacturing
- Traffic control
- Information management
- E-commerce
- Business process management
- Medical domain – monitoring, etc
- Games

Limitations of Agent based solutions

- No overall system controllers
- No global perspective

What is next?

- Understanding AI empowered by heuristic